System Validation Report

Project: Customer Churn Prediction (Team Hackforge)
Course: Software Engineering Mini Project, Deliverable Part 3
Date of test run: 30 October 2026
Generated by: `python tools/generate_test_reports.py` from a single run of the automated test suite

1. Purpose and Scope

This report validates the complete Customer Churn Prediction system against the requirements in SRS v1.1 (`docs/SRS_v1.1_Addendum.md`) by executing every test case in Test Plan v1.1 (`docs/TEST_PLAN_v1.1.md`). The system under test includes the training pipeline, the Streamlit web application, batch scoring, the what-if simulator, and the three-paradigm comparison (classical ML, deep learning and quantum machine learning).

All test cases are automated. The training pipeline and the paradigm benchmark are run end to end in a temporary folder, so the validated production model is never overwritten.

2. Summary

Test cases	Passed	Failed	Not run	Requirements	Requirements validated
26	26	0	0	54	54

3. Test Execution Record

ID	Test case	Requirements	Level	Actual result / evidence	Status
TC-01	Load valid dataset	FR-01, SW-01	Integration	1/1 automated tests passed	Pass
TC-02	Missing file / missing target handled	FR-02, NFR-02, NFR-10, SR-05, SO-03	Unit	7/7 automated tests passed	Pass
TC-03	Target separation and encoding	FR-03, FR-04	Unit	1/1 automated tests passed	Pass
TC-04	XGBoost training with class weighting	FR-05, FR-10	Integration	1/1 automated tests passed	Pass
TC-05	Grid search with stratified cross validation	FR-06, FR-11	Integration	3/3 automated tests passed	Pass
TC-06	Evaluation metrics and reports	FR-07, FR-08, FR-12, NFR-04, UI-01	System	7/7 automated tests passed	Pass
TC-07	Class imbalance handling, test set untouched	FR-10	Integration	3/3 automated tests passed	Pass
TC-08	Model and encoders saved	FR-09, SW-02	System	1/1 automated tests passed	Pass
TC-09	Saved model and encoders reload as a pair	FR-14, NFR-07, SR-03, SO-02	Integration	4/4 automated tests passed	Pass
TC-10	Fresh environment and training runtime	NFR-01, NFR-06, HW-01	Non-functional	3/3 automated tests passed (training seconds = 15.3)	Pass

ID	Test case	Requirements	Level	Actual result / evidence	Status
TC-11	Privacy and repository hygiene	NFR-09, SR-01, SR-02, SR-04, SO-01, COM-01	Non-functional	5/5 automated tests passed	Pass
TC-12	Documentation completeness	FR-13, NFR-05, NFR-08	Non-functional	2/2 automated tests passed	Pass
TC-13	Repeatability	NFR-03	Non-functional	2/2 automated tests passed	Pass
TC-14	Customer profile form and presets	FR-15, UI-02	System	11/11 automated tests passed	Pass
TC-15	Input validation and boundary guards	FR-16, NFR-08, SR-05	Unit	54/54 automated tests passed	Pass
TC-16	Single customer prediction with risk tier	FR-17	Integration	8/8 automated tests passed	Pass
TC-17	Live evaluation and decision threshold tuning	FR-18	System	7/7 automated tests passed	Pass
TC-18	Batch CSV scoring	FR-19	Integration	2/2 automated tests passed	Pass
TC-19	Churn driver explanation and what-if simulator	FR-20	Integration	2/2 automated tests passed	Pass
TC-20	ML vs DML vs QML side-by-side prediction	FR-21, SR-07	System	6/6 automated tests passed	Pass
TC-21	Deep learning model (DML)	FR-22	Unit	9/9 automated tests passed	Pass
TC-22	Quantum machine learning model (QML)	FR-23	Unit	10/10 automated tests passed	Pass
TC-23	Paradigm benchmark and experiment tracking	FR-24, NFR-12, SR-06, SW-03	System	6/6 automated tests passed	Pass
TC-24	Prediction latency	NFR-11	Non-functional	1/1 automated tests passed (median prediction ms = 11.3)	Pass
TC-25	Production model and encoder integrity	NFR-07, SO-02	Integration	3/3 automated tests passed	Pass
TC-26	Static code quality	NFR-13	Non-functional	pylint static analysis (pylint score = 9.82/10)	Pass

4. Requirement Validation Matrix

Requirement	Test cases	Status
FR-01	TC-01	Validated
FR-02	TC-02	Validated
FR-03	TC-03	Validated
FR-04	TC-03	Validated
FR-05	TC-04	Validated
FR-06	TC-05	Validated
FR-07	TC-06	Validated

Requirement	Test cases	Status
FR-08	TC-06	Validated
FR-09	TC-08	Validated
FR-10	TC-04, TC-07	Validated
FR-11	TC-05	Validated
FR-12	TC-06	Validated
FR-13	TC-12	Validated
FR-14	TC-09	Validated
FR-15	TC-14	Validated
FR-16	TC-15	Validated
FR-17	TC-16	Validated
FR-18	TC-17	Validated
FR-19	TC-18	Validated
FR-20	TC-19	Validated
FR-21	TC-20	Validated
FR-22	TC-21	Validated
FR-23	TC-22	Validated
FR-24	TC-23	Validated
NFR-01	TC-10	Validated
NFR-02	TC-02	Validated
NFR-03	TC-13	Validated
NFR-04	TC-06	Validated
NFR-05	TC-12	Validated
NFR-06	TC-10	Validated
NFR-07	TC-09, TC-25	Validated
NFR-08	TC-12, TC-15	Validated
NFR-09	TC-11	Validated
NFR-10	TC-02	Validated
NFR-11	TC-24	Validated
NFR-12	TC-23	Validated
NFR-13	TC-26	Validated
UI-01	TC-06	Validated
UI-02	TC-14	Validated
SW-01	TC-01	Validated
SW-02	TC-08	Validated
SW-03	TC-23	Validated
HW-01	TC-10	Validated
COM-01	TC-11	Validated

Requirement	Test cases	Status
SO-01	TC-11	Validated
SO-02	TC-09, TC-25	Validated
SO-03	TC-02	Validated
SR-01	TC-11	Validated
SR-02	TC-11	Validated
SR-03	TC-09	Validated
SR-04	TC-11	Validated
SR-05	TC-02, TC-15	Validated
SR-06	TC-23	Validated
SR-07	TC-20	Validated

5. Non-functional Measurements

Requirement	Target	Measured	Status
NFR-01 Training time (full params.yaml grid, 3 model families)	< 300 s	15.3 s	Pass
NFR-11 Single prediction latency (median of 30)	< 200 ms	11.3 ms	Pass
NFR-13 Pylint score	>= 8.5/10	9.82/10	Pass
NFR-13 Statement coverage of application modules	>= 85%	89.1%	Pass
NFR-13 Branch coverage of application modules	>= 75%	79.4%	Pass

6. Model Quality Validation

The production XGBoost model was re-evaluated on the 1,409-customer held-out test set during the run (test `test_live_evaluation_matches_saved_benchmark`) and reproduced its recorded benchmark:

Accuracy	Precision (churn)	Recall (churn)	F1 (churn)	ROC-AUC
75.66%	52.92%	75.13%	0.621	0.841

The ML vs DML vs QML benchmark (`paradigm_metrics.json`, the same split for every model) gave the following results:

Paradigm	Model	Accuracy	Recall (churn)	F1 (churn)	ROC-AUC	Parameters
ML	XGBoost	75.66%	79.41%	0.634	0.846	5,628
DML	Deep MLP	73.24%	79.68%	0.613	0.841	5,249
ML	Logistic Regression	73.67%	79.41%	0.616	0.839	20
QML	Quantum VQC	73.74%	75.40%	0.604	0.829	61

7. Conclusion

All 26 test cases passed and all 54 requirements of SRS v1.1 (functional, non-functional, interface and security) are validated. The system is ready for the final demonstration.